

# **Counting, Permutations, & Combinations**

A counting problem asks  
“**how many ways**” some  
event can occur.

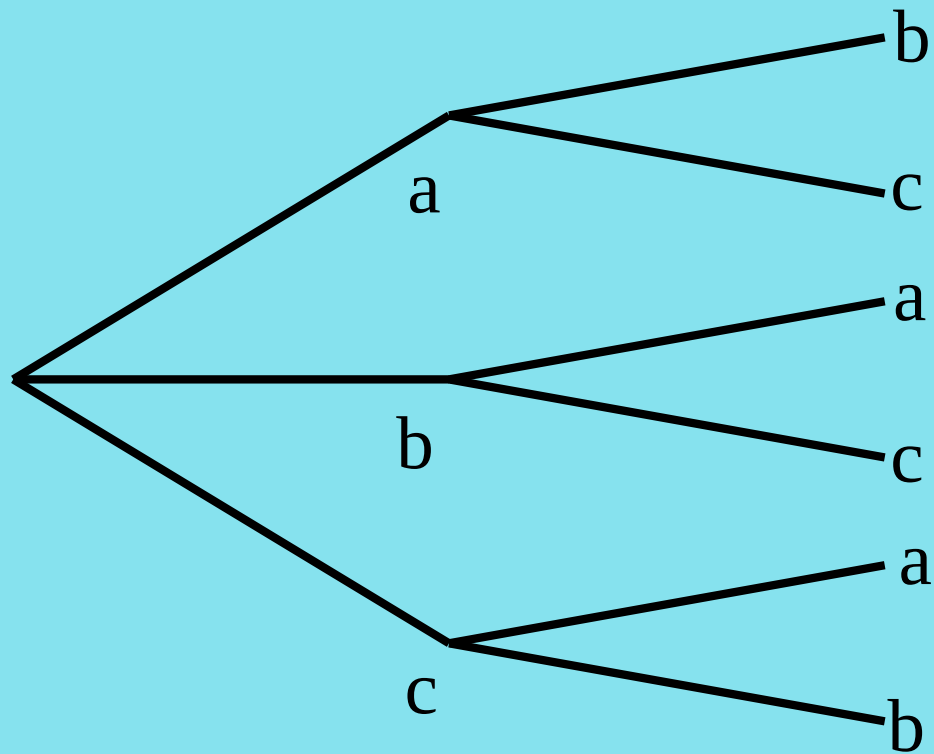
*Ex. 1: How many three-letter codes are there using letters A, B, C, and D if no letter can be repeated?*

- One way to solve is to list all possibilities.

Ex. 2: An experimental psychologist uses a sequence of two food rewards in an experiment regarding animal behavior. These two rewards are of three different varieties. How many different sequences of rewards are there if each variety can be used only once in each sequence?

Next slide

- Another way to solve is a factor tree where the number of end branches is your answer.



# Fundamental Counting Principle

Suppose that a certain procedure  $P$  can be broken into  $n$  successive ordered stages,  $S_1, S_2, \dots, S_n$ , and suppose that

$S_1$  can occur in  $r_1$  ways.

$S_2$  can occur in  $r_2$  ways.

$S_n$  can occur in  $r_n$  ways.

Then the number of ways  $P$  can occur is

$$r_1 \cdot r_2 \cdots r_n$$

Ex. 2: An experimental psychologist uses a sequence of two food rewards in an experiment regarding animal behavior. These two rewards are of three different varieties. How many different sequences of rewards are there if each variety can be used only once in each sequence?

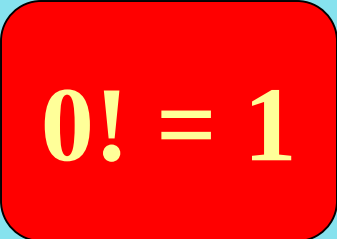
*Using the fundamental counting principle:*

$$\begin{array}{ccc} \underline{3} & \times & \underline{2} \\ \text{1st reward} & & \text{2nd reward} \end{array}$$

# Permutations

An  $r$ -permutation of a set of  $n$  elements is an **ordered** selection of  $r$  elements from the set of  $n$  elements

$${}_n P_r = \frac{n!}{(n-r)!}$$


$$0! = 1$$

Ex. 1: How many three-letter codes are there using letters A, B, C, and D if no letter can be repeated?

*Note: The order does matter*

$${}_4P_3 = \frac{4!}{1!} = 24$$

# Combinations

The number of combinations of  $n$  elements taken  $r$  at a time is

$${}_nC_r = \frac{n!}{r!(n-r)!}$$

Where  $n$  &  $r$  are nonnegative integers &  $r \leq n$

Ex. 3: How many committees of three can be selected from four people?

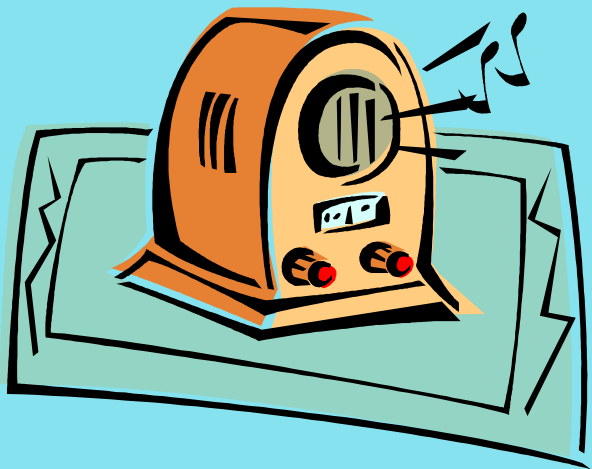
Use A, B, C, and D to represent the people

Note: Does the order matter?

$${}_4C_3 = \frac{4!}{3!1!} = 4$$

Ex. 4: How many ways can the 4 call letters of a radio station be arranged if the first letter must be W or K and no letters repeat?

$$2 \cdot 25 \cdot 24 \cdot 23 = 27,600$$



Ex. 5: In how many ways can our class elect a president, vice-president, and secretary if no student can hold more than one office?

$${}_nP_3 =$$



Ex. 7: Given the digits 5, 3, 6, 7, 8, and 9, how many 3-digit numbers can be made if the first digit must be a prime number? (digits can't be repeated?)

$$3 \cdot 5 \cdot 4 = 60$$

Ex. 8: In how many ways can 9 horses place 1<sup>st</sup>, 2<sup>nd</sup>, or 3<sup>rd</sup> in a race?

$${}_9P_3 = 504$$



Ex. 9: Suppose there are 15 girls and 18 boys in a class. In how many ways can 2 girls and 2 boys be selected for a group of 4 in a project?

$${}_{15}C_2 \times {}_{18}C_2 = 16,065$$



# Permutations

**Definition:** A **permutation** of a set  $S$  is an ordered sequence that contains every element of  $S$  exactly once.

For example, here are all six permutations of the set  $\{a, b, c\}$ :

$(a, b, c)$   $(a, c, b)$   $(b, a, c)$

$(b, c, a)$   $(c, a, b)$   $(c, b, a)$

Ordering is important here.

You can think of a permutation as a ranking of the elements.

How many permutations of an  $n$ -element set are there?

So the above question is asking how many rankings of an  $n$ -element set.

How many permutations of an n-element set are there?

- There are n choices for the first element.
- For each of these, there are n - 1 remaining choices for the second element.
- For every combination of the first two elements, there are n - 2 ways to choose the third element, and so forth.
- Thus, there are a total of

$n \cdot (n - 1) \cdot (n - 2) \cdots 3 \cdot 2 \cdot 1 = n!$  ← This is called n **factorial**.  
permutations of an n-element set.

## Example: Permutation

Suppose each digit is an element in  $\{1,2,3,4,5,6,7,8,9\}$ .

How many 9-digit numbers are there where each nonzero digit appears once?

Each such number corresponds to a permutation of 123456789,  
and each permutation corresponds to such a number.

So the numbers of such numbers is equal to

the number of permutations of  $\{1,2,3,4,5,6,7,8,9\}$ .

Hence there are exactly  $9!$  such numbers.

## **$r$ -permutation**

An ordered arrangement of  $r$  elements of a set is called an  **$r$ -permutation**.

EXAMPLE: Let  $S = \{1, 2, 3\}$ .

The ordered arrangement 3, 1, 2 is a permutation of  $S$ .

The ordered arrangement 3, 2 is a 2-permutation of  $S$ .

The number of  $r$ -permutations of a set with  $n$  elements is denoted by  $P(n, r)$ .

*THEOREM: If  $n$  is a positive integer and  $r$  is an integer with  $1 \leq r \leq n$ , then there are*

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

*$r$ -permutations of a set with  $n$  distinct elements.*

**COROLLARY:** If  $n$  and  $r$  are integers with  $0 \leq r \leq n$ , then

$$P(n, r) = n!/(n - r)! .$$

**EXAMPLE:** How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

**Solution:** Because it matters which person wins which prize, the number of ways to pick the three prize winners is the number of ordered selections of three elements from a set of 100 elements, that is, the number of 3-permutations of a set of 100 elements.

Consequently, the answer is

$$P(100, 3) = 100 \cdot 99 \cdot 98 = 970,200.$$

**EXAMPLE:** How many permutations of the letters  $ABCDEFGH$  contain the string  $ABC$  ?

**Solution:** Because the letters  $ABC$  must occur as a block, we can find the answer by finding the number of permutations of six objects, namely, the block  $ABC$  and the individual letters  $D$ ,  $E$ ,  $F$ ,  $G$ , and  $H$ .

Because these six objects can occur in any order, there are  $6! = 720$  permutations of the letters  $ABCDEFGH$  in which  $ABC$  occurs as a block.

## Combinations

How many subsets of size  $k$  of an  $n$ -element set?

Consider the set  $\{1,2,3,4,5\}$  where  $n=5$ . If  $k=2$ ,

If  $k=2$ , then there are 10 possible subsets of size 2,

i.e.  $\{1,2\}, \{1,3\}, \{1,4\}, \{1,5\}, \{2,3\}, \{2,4\}, \{2,5\}, \{3,4\}, \{3,5\}, \{4,5\}$ .

If  $k=3$

If  $k=3$ , then there are also 10 possible subsets of size 3,

i.e.  $\{1,2,3\}, \{1,2,4\}, \{1,2,5\}, \{1,3,4\}, \{1,3,5\}$

$\{1,4,5\}, \{2,3,4\}, \{2,3,5\}, \{2,4,5\}, \{3,4,5\}$

Ordering is NOT important here.

Think of it as the number of ways to form a committee of  $k$  people from  $n$  people.

## Combinations

How many subsets of size  $k$  of an  $n$ -element set?

- There are  $n$  choices for the first element.
- For each of these, there are  $n - 1$  remaining choices for the second element.
- There are  $n - k + 1$  remaining choices for the last element.
- Thus, there are a total of  
$$n \cdot (n - 1) \cdot (n - 2) \cdots (n - k + 1)$$
to choose  $k$  elements.  
when the ordering is important.

So far we counted the number of ways to choose  $k$  elements,  
when the ordering is important.

e.g.  $\{1,2,3\}$ ,  $\{1,3,2\}$ ,  $\{2,1,3\}$ ,  $\{2,3,1\}$ ,  $\{3,1,2\}$ ,  $\{3,2,1\}$  will be counted as 6 different ways.

## Combinations

How many subsets of size  $k$  of an  $n$ -element set?

How many different ordering of  $k$  elements are (over)-counted?

e.g. If we are forming subsets of size 3, then

$(1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2), (3,2,1)$

are counted as 6 different ways if the ordering is important.

In general, each subset of size  $k$  has  $k!$  different orderings,  
and so each subset is counted  $k!$  times in the above way of choosing  $k$  elements.

• So, when the ordering is not important, the answer is:

This is the  
shorthand for  
"n choose k"

$$\binom{n}{k} = \frac{n \cdot (n-1) \cdots (n-k+1)}{k!} = \frac{n!}{(n-k)!k!}$$

## Example: Team Formation

There are  $m$  boys and  $n$  girls.

How many ways are there to form a team with 3 boys and 3 girls?

There are  $\binom{m}{3}$  choices of 3 boys and  $\binom{n}{3}$  choices for 3 girls.

So by the product rule there are  $\binom{m}{3} \binom{n}{3}$  choices of such a team.

If  $m < 3$  or  $n < 3$ , then the answer should be zero.

## Example: Bit Strings with $k$ Zeros

How many  $n$ -bit sequences contain  $k$  zeros and  $(n - k)$  ones?

We can think of this problem as choosing  $k$  positions

(out of the  $n$  possible positions) and set them to zeroes

and set the remaining positions to ones.

So the above question is asking the number of possible positions of the  $k$  zeros, and the answer is:

$$\binom{n}{k}$$

### Example: Unbalanced Bit Strings

We say a bit string is unbalanced if there are more ones than zeroes or more zeros than ones.

How many  $n$ -bit strings are unbalanced?

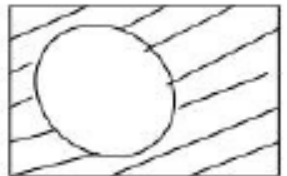
If  $n$  is odd, then every  $n$ -bit string is unbalanced, and the answer is  $2^n$ .

If  $n$  is even, then the number of balanced strings is

$\binom{n}{n/2}$  by choosing  $n/2$  positions to zeroes.

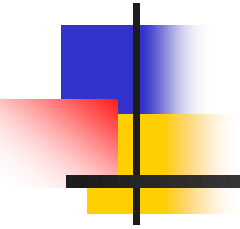
So the number of unbalanced n-bit strings is equal to  
the number of all n-bit strings minus the number of balanced strings,  
and so the answer is  $2^n - \binom{n}{n/2}$

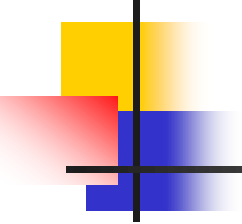
(counting the complement)



# Permutations and Combinations

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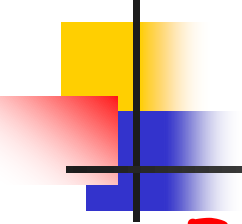




# Objectives:

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- apply fundamental counting principle
- compute permutations
- compute combinations
- distinguish permutations vs combinations

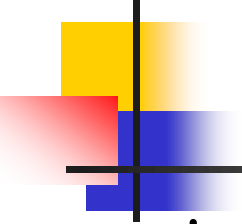


# Fundamental Counting Principle

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**Fundamental Counting Principle** can be used to determine the number of possible outcomes when there are two or more characteristics.

**Fundamental Counting Principle** states that if an event has  $m$  possible outcomes and another independent event has  $n$  possible outcomes, then there are  $m * n$  possible outcomes for the two events together.



# Fundamental Counting Principle

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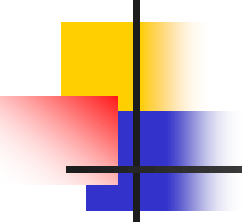
Lets start with a simple example.

A student is to roll a die and flip a coin.  
How many possible outcomes will there be?

|    |    |    |    |    |    |
|----|----|----|----|----|----|
| 1H | 2H | 3H | 4H | 5H | 6H |
| 1T | 2T | 3T | 4T | 5T | 6T |

$6 * 2 = 12$  outcomes

12 outcomes

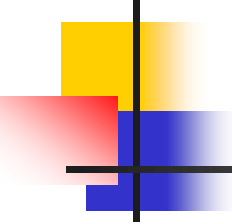


# Fundamental Counting Principle

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For a college interview, Robert has to choose what to wear from the following: 4 slacks, 3 shirts, 2 shoes and 5 ties. How many possible outfits does he have to choose from?

$$4 * 3 * 2 * 5 = 120 \text{ outfits}$$



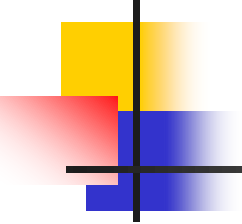
# Permutations

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A **Permutation** is an arrangement of items in a particular order.

Notice, **ORDER MATTERS!**

To find the number of Permutations of  $n$  items, we can use the Fundamental Counting Principle or factorial notation.



# Permutations

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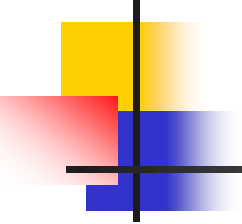
The number of ways to arrange the letters ABC:

|                                     |       |       |       |
|-------------------------------------|-------|-------|-------|
|                                     | _____ | _____ | _____ |
| Number of choices for first blank?  | 3     | _____ | _____ |
| Number of choices for second blank? | 3     | 2     | _____ |
| Number of choices for third blank?  | 3     | 2     | 1     |

$$3*2*1 = 6$$

$$3! = 3*2*1 = 6$$

ABC    ACB    BAC    BCA    CAB    CBA



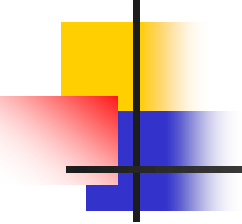
# Permutations

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To find the number of Permutations of  $n$  items chosen  $r$  at a time, you can use the formula

$${}_n P_r = \frac{n!}{(n-r)!} \quad \text{where } 0 \leq r \leq n.$$

$${}_5 P_3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = 5 * 4 * 3 = \mathbf{60}$$



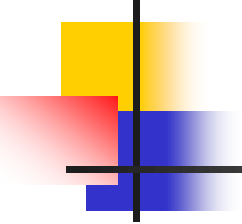
# Permutations

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## Practice:

A combination lock will open when the right choice of three numbers (from 1 to 30, inclusive) is selected. How many different lock combinations are possible assuming no number is repeated?

**Answer Now**



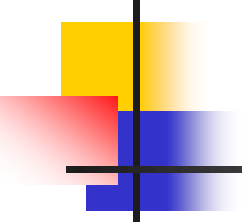
# Permutations

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## Practice:

A combination lock will open when the right choice of three numbers (from 1 to 30, inclusive) is selected. How many different lock combinations are possible assuming no number is repeated?

$${}_{30}P_3 = \frac{30!}{(30-3)!} = \frac{30!}{27!} = 30 * 29 * 28 = 24360$$



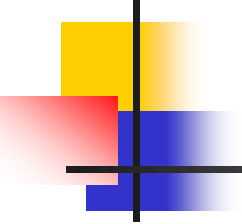
# Permutations

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## Practice:

From a club of 24 members, a President, Vice President, Secretary, Treasurer and Historian are to be elected. In how many ways can the offices be filled?

**Answer Now**



# Permutations

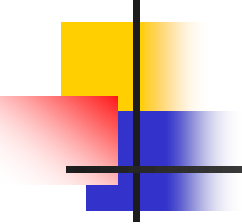
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## Practice:

From a club of 24 members, a President, Vice President, Secretary, Treasurer and Historian are to be elected. In how many ways can the offices be filled?

$${}_{24}P_5 = \frac{24!}{(24-5)!} = \frac{24!}{19!} =$$

$$24 * 23 * 22 * 21 * 20 = 5,100,480$$



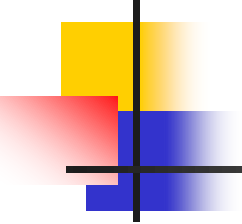
# Combinations

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A **Combination** is an arrangement of items in which order does not matter.

**ORDER DOES NOT MATTER!**

Since the order does not matter in combinations, there are fewer combinations than permutations. The combinations are a "subset" of the permutations.

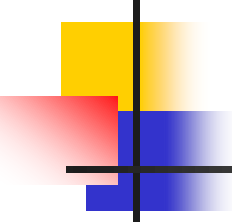


# Combinations

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To find the number of Combinations of  $n$  items chosen  $r$  at a time, you can use the formula

$${}_n C_r = \frac{n!}{r!(n-r)!} \quad \text{where } 0 \leq r \leq n.$$



# Combinations

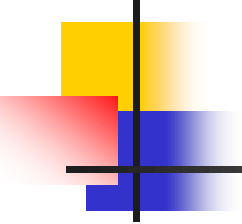
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$${}_n C_r = \frac{n!}{r!(n-r)!} \quad \text{where } 0 \leq r \leq n.$$

$${}_5 C_3 = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} =$$

$$\frac{5 * 4 * 3 * 2 * 1}{3 * 2 * 1 * 2 * 1} = \frac{5 * 4}{2 * 1} = \frac{20}{2} = 10$$



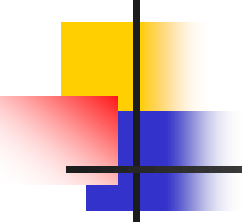
# Combinations

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## Practice:

To play a particular card game, each player is dealt five cards from a standard deck of 52 cards. How many different hands are possible?

**Answer Now**



# Combinations

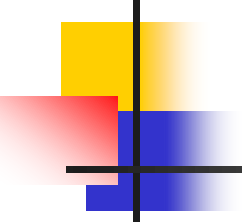
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**Practice:**

To play a particular card game, each player is dealt five cards from a standard deck of 52 cards. How many different hands are possible?

$${}_{52}C_5 = \frac{52!}{5!(52-5)!} = \frac{52!}{5!47!} =$$

$$\frac{52 * 51 * 50 * 49 * 48}{5 * 4 * 3 * 2 * 1} = 2,598,960$$



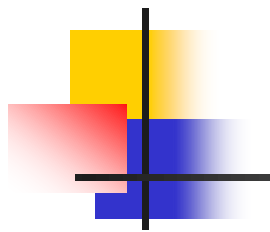
# Combinations

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## Practice:

A student must answer 3 out of 5 essay questions on a test. In how many different ways can the student select the questions?

**Answer Now**

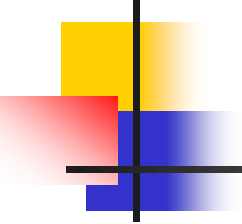


# Combinations

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**Practice:** A student must answer 3 out of 5 essay questions on a test. In how many different ways can the student select the questions?

$${}_5C_3 = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} = \frac{5*4}{2*1} = 10$$



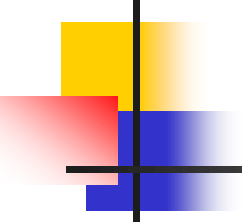
# Combinations

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## Practice:

A basketball team consists of two centers, five forwards, and four guards. In how many ways can the coach select a starting line up of one center, two forwards, and two guards?

**Answer Now**



# Combinations

**Practice:** A basketball team consists of two centers, five forwards, and four guards. In how many ways can the coach select a starting line up of one center, two forwards, and two guards?

Center:

$${}_2C_1 = \frac{2!}{1!1!} = 2$$

Forwards:

$${}_5C_2 = \frac{5!}{2!3!} = \frac{5*4}{2*1} = 10$$

Guards:

$${}_4C_2 = \frac{4!}{2!2!} = \frac{4*3}{2*1} = 6$$

$${}_2C_1 * {}_5C_2 * {}_4C_2$$

Thus, the number of ways to select the starting line up is  $2*10*6 = 120$ .

## THE PIGEONHOLE PRINCIPLE

If  $k$  is a positive integer and  $k + 1$  or more objects are placed into  $k$  boxes, then there is at least one box containing two or more of the objects.

### Example

If  $kn + 1$  (where  $k$  is a positive integer) pigeons are distributed among  $n$  holes then some hole contains at least  $k + 1$  pigeons.

### Applications of Pigeonhole Principle:

This principle is applicable in many fields like Number Theory, Probability, Algorithms, Geometry, etc.

## Problems and Solutions:

### Problem 1

A bag contains beads of two colours: black and white. What is the smallest number of beads which must be drawn from the bag, without looking so that among these beads, two are of the same colour?

Solution: We can draw three beads from bags. If there were no more than one bead of each colour among these, then there would be no more than two beads altogether. This is obvious and contradicts the fact that we have chosen three beads. On the other hand, it is clear that choosing two beads is not enough. Here the beads play the role of pigeons, and the colours (black and white) play the role of pigeonhole.

## Problem 2

Find the minimum number of students in a class such that three of them are born in the same month?

Solution: Number of month  $n = 12$

According to the given condition,

$$K+1 = 3$$

$$K = 2$$

$$M = kn + 1 = 2 \cdot 12 + 1 = 25.$$

Example. A basket of fruit is being arranged out of apples, bananas, and oranges. What is the smallest number of pieces of fruit that should be put in the basket in order to guarantee that either there are at least 8 apples or at least 6 bananas or at least 9 oranges?

Solution:  $8 + 6 + 9 - 3 + 1 = 21$

Example. Pick three natural numbers. Prove that there exist a pair whose sum or difference is divisible by 3.

Solution: Observe that a number when divided by 3 has remainder 0, 1, 2. Identify these remainders as the pigeonholes. So there are 3 pigeonholes.

Given a number it goes to the  $i^{\text{th}}$  pigeonhole if the remainder after dividing it by 3 is  $i$  where  $i = 0, 1$  or  $2$ .

Since there are 3 numbers there can be two cases.

Case(i)

When each number goes to distinct box. Take the numbers in the box 1 and 2 and add them.

Case(ii)

When a box contains at least two numbers. Then take the difference of those two numbers.

## THE GENERALIZED PIGEONHOLE PRINCIPLE

If  $N$  objects are placed into  $k$  boxes, then there is at least one box containing at least  $\lceil N/k \rceil$  objects.

### EXAMPLE

Among 100 people there are at least  $\lceil 100/12 \rceil = 9$  who were born in the same month.

## EXAMPLE

What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F?

Solution: The minimum number of students needed to ensure that at least six students receive the same grade is the smallest integer  $N$  such that  $\lceil N/5 \rceil = 6$ .

The smallest such integer is  $N = 5 \cdot 5 + 1 = 26$ . If you have only 25 students, it is possible for there to be five who have received each grade so that no six students have received the same grade.

Thus, 26 is the minimum number of students needed to ensure that at least six students will receive the same grade.

**EXAMPLE** During a month with 30 days, a baseball team plays at least one game a day, but no more than 45 games. Show that there must be a period of some number of consecutive days during which the team must play exactly 14 games.

**Solution:** Let  $a_j$  be the number of games played on or before the  $j$ th day of the month. Then  $a_1, a_2, \dots, a_{30}$  is an increasing sequence of distinct positive integers, with  $1 \leq a_j \leq 45$ .

Moreover,  $a_1 + 14, a_2 + 14, \dots, a_{30} + 14$  is also an increasing sequence of distinct positive integers, with  $15 \leq a_j + 14 \leq 59$ .

The 60 positive integers  $a_1, a_2, \dots, a_{30}, a_1 + 14, a_2 + 14, \dots, a_{30} + 14$  are all less than or equal to 59. Hence, by the pigeonhole principle two of these integers are equal. Because the integers  $a_j, j = 1, 2, \dots, 30$  are all distinct and the integers  $a_j + 14, j = 1, 2, \dots, 30$  are all distinct, there must be indices  $i$  and  $j$  with  $a_i = a_j + 14$ . This means that exactly 14 games were played from day  $j + 1$  to day  $i$ .

## THEOREM

Every sequence of  $n^2 + 1$  distinct real numbers contains a subsequence of length  $n + 1$  that is either strictly increasing or strictly decreasing.

## EXAMPLE

The sequence 8, 11, 9, 1, 4, 6, 12, 10, 5, 7 contains 10 terms. Note that  $10 = 3^2 + 1$ .

There are four strictly increasing subsequences of length four, namely, 1, 4, 6, 12; 1, 4, 6, 7; 1, 4, 6, 10; and 1, 4, 5, 7.

There is also a strictly decreasing subsequence of length four, namely, 11, 9, 6, 5.

## Pigeonhole Principle

Statement: If  $k$  is a +ve integer &  $k+1$  or more objects are placed into  $k$  boxes, then there is at least one box containing two or more of the objects.

Note: If  $kn+1$  (where  $k$  is a +ve integer) pigeons are distributed among  $n$  holes then some hole contains at least  $k+1$  pigeons.

Prob: Find the min no. of students in a class s.t. three of them were born in the same month. (pigeonholes)

Sol: No. of month = 12, by given cond:  $k+1 = 3 \Rightarrow k = 2$ .  $[2.5] = 2$ .  
min no of students =  $kn+1 = 2 \times 12 + 1 = \boxed{25}$   $[4.5] = 5$ .

### Generalized pigeonhole principle

If  $k$  pigeons are accommodated in  $n$  pigeonholes &  $k > n$  then one of the pigeonholes must contain at least  $\left\lceil \frac{k-1}{n} \right\rceil + 1$  pigeons.  $\left\lceil \frac{k}{n} \right\rceil$   $[5.5] = 6$

Problem: If seven colours are used to paint 50 bicycles then show that at least 8 bicycles will be of same colour.

Sol: No of pigeon =  $k$  = no. of cycles = 50  
No. of holes =  $n$  = no. of colours = 7

By pigeonhole principle we have  $\left\lceil \frac{50-1}{7} \right\rceil + 1 = 8$

$\therefore$  at least 8 bicycles will be of same colour.

Problem: If we select any group of 1000 students on campus, show that at least three of them must have same birthday.

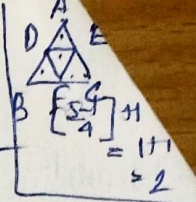
Sol: Max. no. of days in a year = 366  
No of students = no. of pigeons =  $k = 1000$ .

No of days in a year = no. of holes =  $n = 366$ .

By GHP: min. no. of students having same birthday

$$\left\lceil \frac{1000-1}{366} \right\rceil + 1 = 3$$

Prob: Prove that in an equilateral triangle, whose sides are of length 1 unit, if any five pts are chosen, then at least two of them lies in a triangle whose sides are less than  $\frac{1}{2}$



Sol: Let D, E, F are mid pts of the side AB, BC & AC resp. so that the  $\triangle ABC$  divided into four equilateral  $\triangle$ s of each side  $\frac{1}{2}$

No. of pigeons = No. of interior pts = 5.

No. of pigeonholes = No. of  $\triangle$ 's = 4.

$\left[\frac{5-1}{4}\right] + 1 = 2$ , or less two pts lies in a  $\triangle$ .

Prob: Prove that  $\exists$  a true integer n s.t. m divides  $2^n - 1$ , where m being a true odd integer.

Proof: Consider (m+1) true integers,  $2^1 - 1, 2^2 - 1, 2^3 - 1, \dots, 2^m - 1, 2^{m+1} - 1$

By the PHP, when these are divided by m two of the nos will give the same remainder. [These m+1 numbers are the  $\rightarrow$  m+1 pigeons & m remainders  $\rightarrow 0, 1, \dots, m-1$  are m pigeonholes]

Let the two nos be  $2^s - 1$  &  $2^t - 1$  which give same remainder r, when division by m,  $1 \leq s < t \leq m+1$

$$\therefore 2^s - 1 = q_1 m + r \quad \& \quad 2^t - 1 = q_2 m + r$$

$$2^s - 2^t = (q_1 - q_2)m$$

$$\text{But } 2^s - 2^t = 2^s (2^{t-s} - 1)$$

$$\therefore 2^s (2^{t-s} - 1) = (q_1 - q_2)m, \quad \text{but } m = \text{odd, \& hence cannot be factor of } 2^s.$$

$$\therefore m \mid 2^{t-s} - 1 \quad \text{Take } t-s = n \quad \therefore m \mid 2^n - 1.$$

If  $n$  pigeon holes are occupied by  $(kn+1)$  pigeons,  $k$  is a +ve integer, prove that at least one pigeon hole is occupied by  $k+1$  or more pigeons.

Hence find the min no. of integers to be selected from

$S = \{1, 2, \dots, 9\}$  so that the sum of the integers are even.

Proof: Assume  $n$  holes are occupied by  $kn+1$  pigeons.

Claim: At least one hole is occupied by  $k+1$  or more pigeons.

If not, at least one hole is not occupied by  $k+1$  or more "

$\Rightarrow$  each hole contains at most  $k$  pigeons.

Since we have  $n$  holes then the total no. of pigeon  $\leq kn. \Rightarrow \Leftarrow$ .

Sol<sup>n</sup>: Let us divide set  $S = \{1, 2, \dots, 9\}$  in two subsets  $\{1, 3, 5, 7, 9\}$  &  $\{2, 4, 6, 8\}$  which may be treated as pigeon holes.  $\therefore n=2$

At least two nos must be chosen either from the first subset / from the 2nd.

i.e. at least one pigeonhole contains two pigeons.

$$\text{i.e. } kn+1 = 2 \Rightarrow k=1$$

$\therefore$  the <sup>min</sup> no. of the pigeon required / min no. of integers to be selected is equal to  $kn+1 = 1 \cdot 2 + 1 = \boxed{3}$

~~1, 2, 3, 4, 5, 6, 7, 8, 9~~

## The Subtraction Rule (Principle of Inclusion–Exclusion)

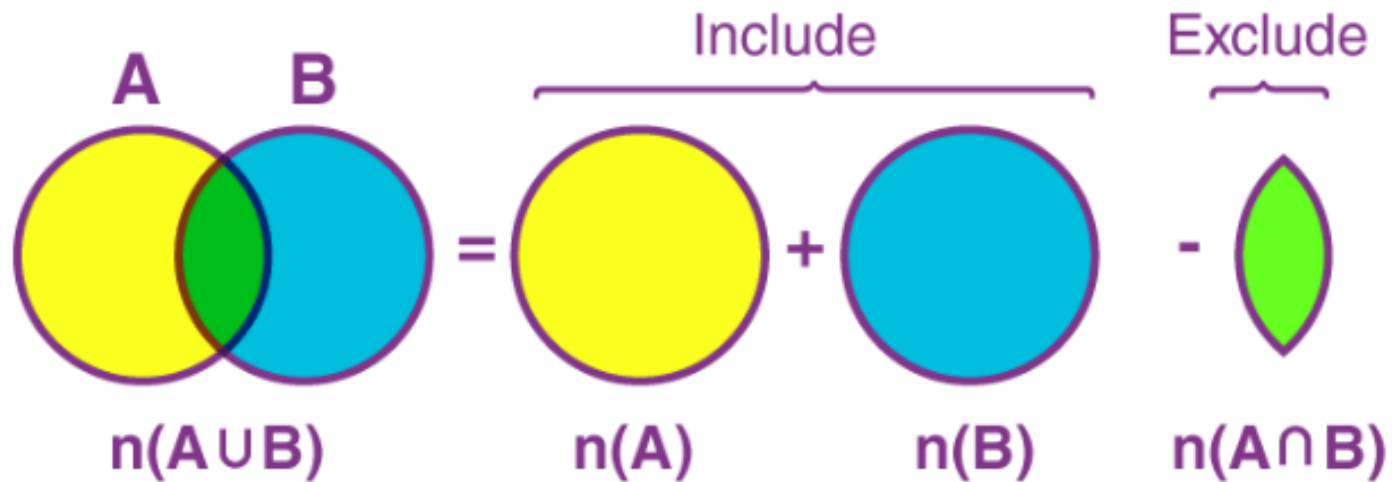
If a task can be done in either  $n_1$  ways or  $n_2$  ways, then the number of ways to do the task is  $n_1 + n_2$  minus the number of ways to do the task that are common to the two different ways.

Principle of Inclusion and Exclusion is an approach which derives the method of finding the number of elements in the union of two finite sets. This is used for solving combinations and probability problems when it is necessary to find a counting method, which makes sure that an object is not counted twice.

Consider two finite sets A and B. We can denote the Principle of Inclusion and Exclusion formula as follows.

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

# The Subtraction Rule (Inclusion–Exclusion for Two Sets)



## Example :

Among 50 patients admitted to a hospital, 25 are diagnosed with pneumonia, 30 with bronchitis, and 10 with both pneumonia and bronchitis. Determine:

(a) The number of patients diagnosed with pneumonia or bronchitis (or both).

(b) The number of patients not diagnosed with pneumonia or bronchitis.

## Solution:

The first step is to formally identify the sets and indicate the number of elements in each. This can be done purely with the given information; No calculation is necessary. With this inclusion-exclusion principle question, the three sets can be defined as follows:

Let  $U$  denote the entire set of patients. Let  $P$  and  $B$  denote the set of patients diagnosed with pneumonia and bronchitis respectively.

Thus:  $|U| = 50$ ,  $|P| = 25$ ,  $|B| = 30$ ,  $|P \cap B| = 10$ .

Now we can calculate how many elements live only in  $P$  but not  $|P \cap B|$ :  
Since  $|P| = 25$  and  $|P \cap B| = 10$ , there are 15 ( $25 - 10 = 15$ ) elements exclusive to  $P$ .

**Solution cont. :**

Follow the same method to calculate the number of elements living only in B but not  $|P \cap B|$ :

Since  $|B| = 30$  and  $|P \cap B| = 10$ , there are 20 ( $30 - 10 = 20$ ) elements exclusive to B.

(a) By using the inclusion-exclusion principle:

$$\begin{aligned}|P \cup B| &= |P| + |B| - |P \cap B| \\ &= (25 + 30) - (10) \\ &= 45\end{aligned}$$

Thus 45 patients are diagnosed with pneumonia or bronchitis.

(b) This is the same as asking to determine  $|(P \cup B)'|$ . We know that there are 50 patients altogether – of which 45 are diagnosed with pneumonia or bronchitis. Use this to solve the question:

$$|U| = 50.$$

$$|P \cup B| = 45$$

Therefore,

$$\begin{aligned}|(P \cup B)'| \\ &= 50 - 45 = 5\end{aligned}$$

5 patients are not diagnosed with pneumonia or bronchitis.

## Example :

In a discrete mathematics class every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class?

### Solution:

Let  $A$  be the set of students in the class majoring in computer science and  $B$  be the set of students in the class majoring in mathematics. Then  $A \cap B$  is the set of students in the class who are joint mathematics and computer science majors. Because every student in the class is majoring in either computer science or mathematics (or both), it follows that the number of students in the class is  $|A \cup B|$ . Therefore,

$$|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30.$$

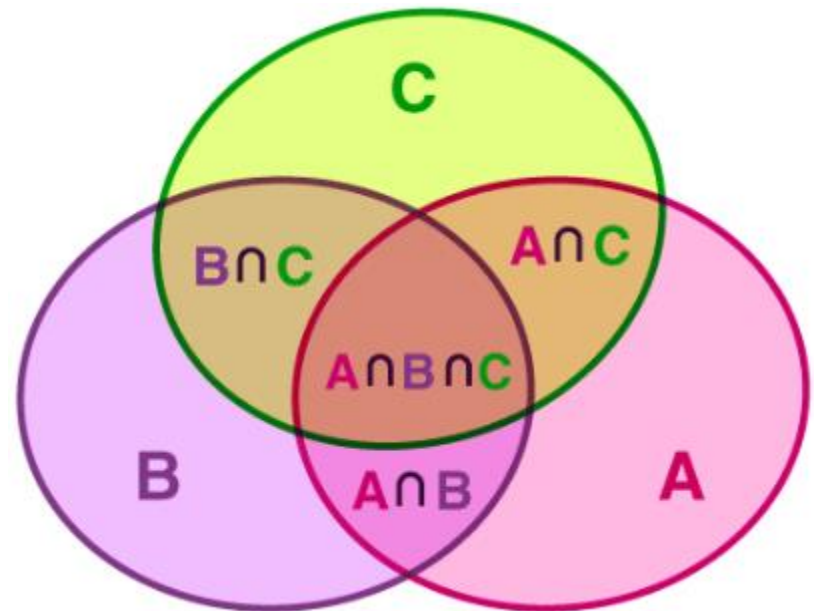
Therefore, there are 30 students in the class.

## The Subtraction Rule (Inclusion–Exclusion for Three Sets)

If we have 3 sets A, B, and C, then according to the Principle of Inclusion and Exclusion,

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Principle of Inclusion and Exclusion represented by a Venn diagram for 3 sets is given below.



## Example :

Among a group of students, 49 study Physics, 37 study English and 21 study Biology. If 9 of these students study Physics and English, 5 study English and Biology, 4 study Physics and Biology and 3 study Physics, English and Biology, find the number of students in the group.

### Solution:

Let P represent the number of students who study Physics, E represent the number of students who study English and B represent the number of students who study Biology.

Number of students in the group =  $n(P \cup E \cup B)$

Given  $n(P) = 49$ ,  $n(E) = 37$ ,  $n(B) = 21$

$$n(P \cap E) = 9$$

$$n(E \cap B) = 5$$

$$n(P \cap B) = 4$$

$$n(P \cap E \cap B) = 3$$

$$\begin{aligned} n(P \cup E \cup B) &= n(P) + n(E) + n(B) - n(P \cap E) - n(E \cap B) - n(P \cap B) + n(P \cap E \cap B) \\ &= 49 + 37 + 21 - 9 - 5 - 4 + 3 \\ &= 92 \end{aligned}$$

## Example :

A large software development company employs 100 computer programmers. Of them, 45 are proficient in Java, 30 in C#, 20 in Python, six in C# and Java, one in Java and Python, five in C# and Python, and just one programmer is proficient in all three languages above.

### Solution:

Let  $U$  denote the set of all employed computer programmers and let  $J$ ,  $C$  and  $P$  denote the set of programmers proficient in Java, C# and Python, respectively.

Thus:  $|U| = 100$ ,  $|J| = 45$ ,  $|C| = 30$ ,  $|P| = 20$ ,  $|J \cap C| = 6$ ,  $|J \cap P| = 1$ ,  $|C \cap P| = 5$ ,  $|J \cap C \cap P| = 1$

We need to determine the cardinality of the complement of the set  $J \cup C \cup P$ . (This is denoted as  $|(J \cup C \cup P)'|$ ).

Calculate  $|J \cup C \cup P|$  first before determining the complement value:

$$|J \cup C \cup P| = 39 + 5 + 20 + 4 + 15 + 1 = 84$$

Now calculate the complement:

$$|(J \cup C \cup P)'| = |U| - |J \cup C \cup P| = 100 - 84 = 16$$

16 programmers are not proficient in any of the three languages.

**THEOREM 1**

**THE PRINCIPLE OF INCLUSION-EXCLUSION** Let  $A_1, A_2, \dots, A_n$  be finite sets. Then

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| = & \sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ & + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|. \end{aligned}$$

**EXAMPLE 5** Give a formula for the number of elements in the union of four sets.

**Solution:** The inclusion–exclusion principle shows that

$$\begin{aligned} |A_1 \cup A_2 \cup A_3 \cup A_4| = & |A_1| + |A_2| + |A_3| + |A_4| \\ & - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_1 \cap A_4| - |A_2 \cap A_3| - |A_2 \cap A_4| \\ & - |A_3 \cap A_4| + |A_1 \cap A_2 \cap A_3| + |A_1 \cap A_2 \cap A_4| + |A_1 \cap A_3 \cap A_4| \\ & + |A_2 \cap A_3 \cap A_4| - |A_1 \cap A_2 \cap A_3 \cap A_4|. \end{aligned}$$

Note that this formula contains 15 different terms, one for each nonempty subset of  $\{A_1, A_2, A_3, A_4\}$ .

## EXAMPLE

How many positive integers not exceeding 1000 are divisible by 7 or 11?

### Solution:

Let  $A$  be the set of positive integers not exceeding 1000 that are divisible by 7, and let  $B$  be the set of positive integers not exceeding 1000 that are divisible by 11.

Then  $A \cup B$  is the set of integers not exceeding 1000 that are divisible by either 7 or 11, and  $A \cap B$  is the set of integers not exceeding 1000 that are divisible by both 7 and 11.

We know that among the positive integers not exceeding 1000 there are  $1000/7$  integers divisible by 7 and  $1000/11$  divisible by 11. Because 7 and 11 are relatively prime, the integers divisible by both 7 and 11 are those divisible by  $7 \cdot 11$ .

Consequently, there are  $1000/(11 \cdot 7)$  positive integers not exceeding 1000 that are divisible by both 7 and 11. It follows that there are

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= \left\lfloor \frac{1000}{7} \right\rfloor + \left\lfloor \frac{1000}{11} \right\rfloor - \left\lfloor \frac{1000}{7 \cdot 11} \right\rfloor \\ &= 142 + 90 - 12 = 220 \end{aligned}$$

positive integers not exceeding 1000 that are divisible by either 7 or 11.

EXERCISE: Suppose that there are 1807 freshmen at your school. Of these, 453 are taking a course in computer science, 567 are taking a course in mathematics, and 299 are taking courses in both computer science and mathematics. How many are not taking a course either in computer science or in mathematics?

EXERCISE: A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French, and Russian, how many students have taken a course in all three languages?

NOW IT'S TIME FOR...

# ADVANCED COUNTING TECHNIQUES

## Recurrence Relations

- Sequences are ordered lists of elements.
  - 1, 2, 3, 5, 8
  - 1, 3, 9, 27, 81, ...
- Sequences arise throughout mathematics, computer science, and in many other disciplines, ranging from botany to music.
- We will introduce the terminology to represent sequences.

## Recurrence Relations

**Definition:** A **sequence** is a function from a subset of the integers (usually either the set  $\{0, 1, 2, 3, 4, \dots\}$  or  $\{1, 2, 3, 4, \dots\}$ ) to a set  $S$ .

- The notation  $a_n$  is used to denote the image of the integer  $n$ . We can think of  $a_n$  as the equivalent of  $f(n)$  where  $f$  is a function from  $\{0, 1, 2, \dots\}$  to  $S$ . We call  $a_n$  a **term** of the sequence.

**Example:** Consider the sequence  $\{a_n\}$  where  $a_n = 1/n$ .  
Then

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

## Recurrence Relations

**Definition:** A **recurrence relation** for the sequence  $\{a_n\}$  is an equation that expresses  $a_n$  in terms of one or more of the previous terms of the sequence, namely,  $a_0, a_1, \dots, a_{n-1}$ , for all integers  $n$  with  $n \geq n_0$ , where  $n_0$  is a nonnegative integer.

- A sequence is called a **solution** of a recurrence relation if its terms satisfy the recurrence relation.
- The **initial conditions** for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

## Recurrence Relations

**Example:** Let  $\{a_n\}$  be a sequence that satisfies the recurrence relation  $a_n = a_{n-1} + 3$  for  $n = 1, 2, 3, 4, \dots$  and suppose that  $a_0 = 2$ . What are  $a_1$ ,  $a_2$  and  $a_3$ ?

[Here the initial condition is  $a_0 = 2$ .]

**Example:** Let  $\{a_n\}$  be a sequence that satisfies the recurrence relation  $a_n = a_{n-1} - a_{n-2}$  for  $n = 2, 3, 4, \dots$  and suppose that  $a_0 = 3$  and  $a_1 = 5$ . What are  $a_2$  and  $a_3$ ?

[Here the initial conditions are  $a_0 = 3$  and  $a_1 = 5$ .]

## Fibonacci Numbers

**Definition:** the Fibonacci sequence,  $f_0, f_1, f_2, \dots$ , is defined by:

- Initial Conditions:  $f_0 = 0, f_1 = 1$
- Recurrence Relation:  $f_n = f_{n-1} + f_{n-2}$

**Example:** Find  $f_4, f_5, f_6, f_7$  and  $f_8$ .

## Recurrence Relations

- Finding a formula for the  $n$ th term of the sequence generated by a recurrence relation is called **solving the recurrence relation**.
- Such a formula is called a **explicit (closed) formula**.
- One technique for finding an explicit formula for the sequence defined by a recurrence relation is **backtracking**.

## Recurrence Relations

**Backtracking:** From the  $n$ th term equation given which consists of lower terms in the sequence the explicit formula is obtained. The lower terms are substituted in much lower terms and finally all terms on right are expressed in terms of  $n$ .

**Solve:** Find explicit formula for the sequence defined by,

$$c_n = 2c_{n-1} + 1, c_1 = 7$$

**Solution:** The definition of the previous term is substituted in the formula.

## Recurrence Relations

$$\begin{aligned}c_n &= 2c_{n-1} + 1 \\&= 2(2c_{n-2} + 1) + 1 \\&= 2[2(2c_{n-3} + 1) + 1] + 1 \\&= 2^3c_{n-3} + 4 + 2 + 1 \\&= 2^3c_{n-3} + 2^2 + 2^1 + 1\end{aligned}$$

Slowly a regular pattern is coming up.this backtracking ends at

$$\begin{aligned}c_n &= 2^{n-1}c_{n-(n-1)} + 2^{n-2} + 2^{n-3} + ..... + 2^2 + 2^1 + 1 \\&= 2^{n-1}c_1 + 2^{n-1} - 1 \\&= 7.2^{n-1} + 2^{n-1} - 1 \\&= 8.2^{n-1} - 1 \\&= 2^{n+2} - 1\end{aligned}$$

## Recurrence Relations

**Example:** Let  $\{a_n\}$  be a sequence that satisfies the recurrence relation  $a_n = a_{n-1} + 3$  for  $n = 2, 3, 4, \dots$  and suppose that  $a_1 = 2$ . Find an explicit formula for the sequence.

$$\begin{aligned}a_n &= a_{n-1} + 3 \\&= (a_{n-2} + 3) + 3 = a_{n-2} + 2 \cdot 3 \\&= (a_{n-3} + 3) + 2 \cdot 3 = a_{n-3} + 3 \cdot 3 \\&\dots \\&= a_2 + (n - 2) \cdot 3 = (a_1 + 3) + (n - 2) \cdot 3 \\&= a_1 + (n - 1) \cdot 3 = 2 + (n - 1) \cdot 3 \\&= 3 \cdot n - 1\end{aligned}$$

## Recurrence Relations

**Example 1:**  $b_n = 5b_{n-1} + 3, b_1 = 3.$

**Example 2:**  $c_n = c_{n-1} + n, b_1 = 4.$

**Example 3:**  $d_n = n \cdot d_{n-1}, d_1 = 6.$

**Note:** Backtracking can sometimes be cumbersome and lead to equations which are very hectic to handle. Next a very easy method of solving recursion relation.

# Linear Homogeneous Recurrence Relations

**Definition:** A linear homogeneous recurrence relation of degree  $k$  with constant coefficients is a recurrence relation of the form

$$a_n = r_1 a_{n-1} + r_2 a_{n-2} + \cdots + r_k a_{n-k}$$

where  $r_1, r_2, \dots, r_k$  are real numbers, and  $r_k \neq 0$ .

- it is **linear** because the right-hand side is a sum of the previous terms of the sequence each multiplied by a function of  $n$ .
- it is **homogeneous** because no terms occur that are not multiples of the  $a_j$ s. Each coefficient is a **constant**.
- the **degree** is  $k$  because  $a_n$  is expressed in terms of the previous  $k$  terms of the sequence.

## Linear Homogeneous Recurrence Relations

- $p_n = (1.11)p_{n-1}$  linear homogeneous recurrence relation of degree 1
- $f_n = f_{n-1} + f_{n-2}$  linear homogeneous recurrence relation of degree 2
- $a_n = a_{n-1} + a_{n-2}^2$  not linear
- $h_n = 2h_{n-1} + 1$  not homogeneous
- $b_n = nb_{n-1}$  coefficients are not constants

## Linear Homogeneous Recurrence Relations

- The basic approach is to look for solutions of the form  $a_n = x^n$ , where  $x$  is a constant.
- Note that  $a_n = x^n$  is a solution to the recurrence relation

$$a_n = r_1 a_{n-1} + r_2 a_{n-2} + \cdots + r_k a_{n-k}$$

if and only if

$$x^n = r_1 x^{n-1} + r_2 x^{n-2} + \cdots + r_k x^{n-k}.$$

- Algebraic manipulation yields the **characteristic equation**:

$$x^k - r_1 x^{k-1} - r_2 x^{k-2} - \cdots - r_k = 0$$

## Linear Homogeneous Recurrence Relations

- The sequence  $\{a_n\}$  with  $a_n = x^n$  is a **solution** if and only if  **$x$  is a solution to the characteristic equation**.

**Theorem:** If the characteristic equation

$$x^2 - r_1x - r_2 = 0$$

of the recurrence relation  $a_n = r_1a_{n-1} + r_2a_{n-2}$  has two distinct roots,  $x_1$  and  $x_2$ , then

$$a_n = px_1^n + qx_2^n$$

where  $p$  and  $q$  depend on the initial conditions, is the **explicit formula** for the sequence.

## Linear Homogeneous Recurrence Relations

**Theorem:** If the characteristic equation

$$x^2 - r_1x - r_2 = 0$$

of the recurrence relation  $a_n = r_1a_{n-1} + r_2a_{n-2}$  has a single root,  $x$ , then

$$a_n = px^n + qnx^n$$

where  $p$  and  $q$  depend on the initial conditions, is the **explicit formula** for the sequence.

## Linear Homogeneous Recurrence Relations

**Example 1:** Find an explicit formula for the sequence defined by  $a_n = 4a_{n-1} + 5a_{n-2}$  with the initial conditions  $a_1 = 2$  and  $a_2 = 6$ .

**Example 2:** Find an explicit formula for the sequence defined by  $b_n = -6b_{n-1} - 9b_{n-2}$  with the initial conditions  $b_1 = 2.5$  and  $b_2 = 4.7$ .

**Example 3 (Fibonacci sequence):** Find an explicit formula for the sequence defined by  $f_n = f_{n-1} + f_{n-2}$  with the initial conditions  $f_0 = 0$  and  $f_1 = 1$ .

# Generating Function and it's application

$$1 + x + x^2 + x^3 + \dots$$

Polynomial:  $a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

$$1 + x + x^2 + \dots + \dots = \frac{1}{1-x}, \quad |x| < 1$$

$$\frac{d}{dx} \left( \frac{1}{1-x} \right) = 1 + 2x + 3x^2 + \dots + \dots$$

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots$$

$\frac{1}{1-x}$  generates  $1, 1, 1, \dots, 1, \dots$

$$\boxed{\frac{1}{(1-x)^2}}$$

generates  $1, 2, 3, 4, \dots$  and so on.

generating Function of  $1, 2, 3, 4, \dots$

□ What is the generating func<sup>n</sup> which generates  
 $1^2, 2^2, 3^2, 4^2, \dots$

$$\frac{x}{(1-x)^2} = x + 2x^2 + 3x^3 + 4x^4 + \dots$$

$$\frac{d}{dx} \left[ \frac{x}{(1-x)^2} \right] = 1 + 4x + 3^2 x^2 + 4^2 x^3 + \dots$$

$$\Rightarrow \boxed{\frac{x+1}{(1-x)^3}} = 1 + 2^2 x + 3^2 x^2 + 4^2 x^3 + \dots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$x = 2y$$

Then  $\frac{1}{1-2y} = 1 + 2y + 2^2y^2 + 2^3y^3 + \dots$

and  $\frac{1}{1-2y}$  generates  $2^0, 2^1, 2^2, 2^3, 2^4, \dots$

$$x = 3y$$

$\frac{1}{1-3y}$  generates  $3^0, 3^1, 3^2, 3^3, 3^4, \dots$

Defn: The generating function for the sequence .

$a_0, a_1, a_2, \dots, a_k, \dots$  of real numbers is the infinite series

$$G(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{k=0}^{\infty} a_kx^k$$

□ Use the method of generating function to solve the recurrence reln

$$a_n = 3a_{n-1} + 1 \quad n \geq 1$$

given that  $a_0 = 1$

Soln: Let the generating function of  $\{a_n\}$

be given by

$$G(x) = \sum_{n=0}^{\infty} a_n x^n$$

The given recurrence rel<sup>n</sup> is :

$$a_n = 3a_{n-1} + 1$$

$$\Rightarrow \sum_{n=1}^{\infty} a_n x^n = 3 \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} x^n$$

multiplying both sides by  $x^n$  and summing up

$$\Rightarrow G(x) - a_0 = 3x G(x) + \frac{x}{1-x}$$

$$\Rightarrow (1-3x) G(x) = 1 + \frac{x}{1-x}$$

$$= \frac{1}{1-x}$$

$$\Rightarrow G(x) = \frac{1}{(1-x)(1-3x)}$$

$$= -\frac{1}{2} \cdot \frac{1}{(1-x)} + \frac{3}{2} \cdot \frac{1}{(1-3x)}$$

$$= -\frac{1}{2} (1-x)^{-1} + \frac{3}{2} (1-3x)^{-1}$$

$$\Rightarrow \sum_{n=0}^{\infty} a_n x^n = -\frac{1}{2} \sum_{n=0}^{\infty} x^n + \frac{3}{2} \sum_{n=0}^{\infty} 3^n x^n$$

$$\Rightarrow a_n = \text{coefficient of } x^n \text{ in } G(x)$$

$$= \frac{3}{2} [3^n] - \frac{1}{2}$$

$$= \frac{1}{2} (3^{n+1} - 1)$$

□ Prob: Use the method of generating function to solve the recurrence rel<sup>n</sup>:

$$a_n = 4a_{n-1} - 4a_{n-2} + 4^n, \quad n \geq 2$$

$$a_0 = 2, \quad a_1 = 8$$

Sol<sup>n</sup>: Let the generating func<sup>n</sup> of  $\{a_n\}$  be

$$G(x) = \sum_{n=0}^{\infty} a_n x^n$$

Multiplying both sides of the given recurrence relation by  $x^n$  and summing up, we have

$$\sum_{n=2}^{\infty} a_n x^n = 4 \sum_{n=2}^{\infty} a_{n-1} x^n - 4 \sum_{n=2}^{\infty} a_{n-2} x^n + \sum_{n=2}^{\infty} 4^n x^n$$

$$\Rightarrow \{G(x) - a_0 - a_1x\} = 4x \{G(x) - a_0\} - 4x^2 G(x) + \frac{1}{1-4x} - 1 - 4x$$

$$\Rightarrow G(x) \{1 - 4x + 4x^2\} = \frac{1}{1-4x} - 1 - 4x - 8x + 2 + 8x$$

$$= \frac{1}{1-4x} + 1 - 4x$$

$$\Rightarrow G(x) = \frac{1 + (1-4x)^2}{(1-4x)(1-2x)^2}$$

$$= \frac{4}{1-4x} - \frac{2}{(1-2x)^2}$$

Then

$$G(x) = 4 [1 + 4x + (4x)^2 + \dots + (4x)^n] - 2 [1 + 2(2x) + 3(2x)^2 + \dots + (1+n)(2x)^n + \dots]$$

Then,

$$a_n = 4 \cdot 4^n - (2n+2)^n (2^n)$$

$$= 4^{n+1} - 2^{n+1} (n+1)$$

## Problems

1. Use method of generating function to solve the recurrence relation  $a_n = 3a_{n-1} + 1$ ;  $n \geq 1$  given that  $a_0 = 1$ .
2. Solve the recurrence relation  $a_n - 7a_{n-1} + 10a_{n-2} = 0$  for  $n \geq 2$  given that  $a_0 = 10$ ,  $a_1 = 41$  using generating functions.
3. Using generating function solve the recurrence relation corresponding to the Fibonacci sequence,  $a_n = a_{n-1} + a_{n-2}$ ,  $n \geq 2$  with  $a_0 = 1, a_1 = 1$ .
4. Solve the recurrence relation  $S(n+1) - 2S(n) = 4^n$ ,  $S(0) = 1$ ,  $n \geq 0$ .